우주비행학 (Space Flight Dynamics)

2018년 봄학기

Home Work 1

due date: 2018년 3월 29일

문제 1: Problem 1 in Chapter 1

문제 2: Problem 2 in Chapter 1

문제 3: Problem 4 in Chapter 1

문제 4: Problem 9 in Chapter 1

문제 5: Problem 5 in Chapter 1

문제 6: Problem 6 in Chapter 1

문제 7: Problem 10 in Chapter 1

* The number of problem is referred by 2^{nd} edition of the test book.

1.16 PROBLEMS

1. Show that the two expressions (1.17) and (1.18) for the velocity of the astronaut in the space station are equivalent by constructing the rotation matrix R^{si} and transforming one expression into the other.

2. A reentry vehicle is flying through the atmosphere at velocity $\mathbf{V} = V\mathbf{b}_1$ at flight-path angle γ as shown in Figure 1.17. The vehicle keeps its velocity aligned with $\mathbf{b}_1$ as γ changes. Use the velocity rule to show that the inertial acceleration of the center of mass can be written as

$$\mathbf{A} = \dot{V}\,\mathbf{b}_1 + V\dot{\gamma}\,\mathbf{b}_2 \tag{1.141}$$

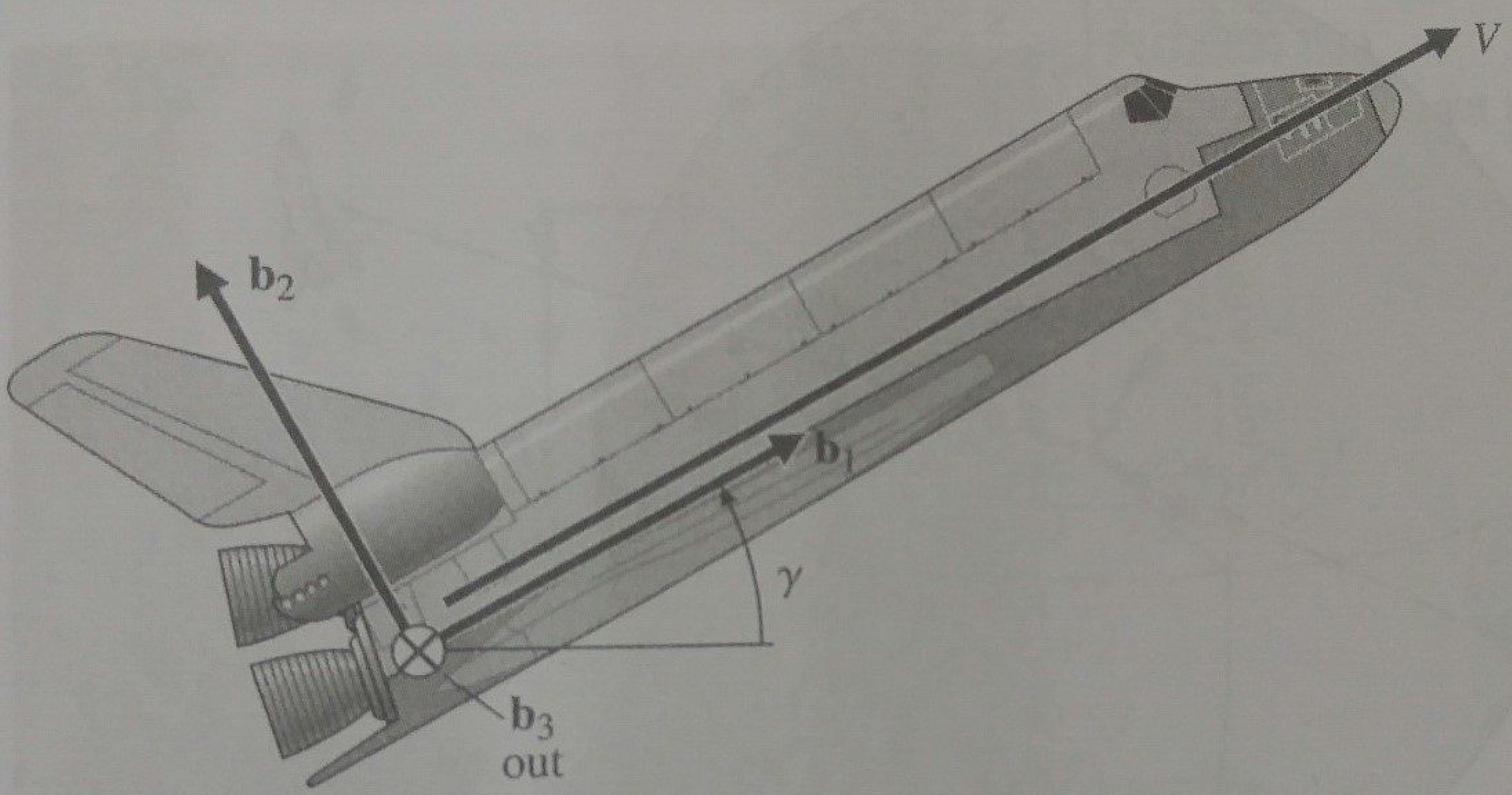

FIGURE 1.17
Acceleration calculation for a reentry vehicle.

3. A rotation matrix $R^{\mathbf{si}}$ can be thought of as a column matrix of the three unit vectors of the $\mathbf{s}$-frame written in their $\mathbf{i}$-frame components

$$R^{\mathbf{si}} = [\mathbf{s}_1 \;\vdots\; \mathbf{s}_2 \;\vdots\; \mathbf{s}_3] \tag{1.142}$$

Use the velocity rule to show that a rotation matrix obeys the differential equation

$$\frac{{}^{\mathbf{i}}d}{dt} R^{\mathbf{si}} = \omega^{\mathbf{si}} \times R^{\mathbf{si}} \tag{1.143}$$

where the cross product is interpreted as the cross product of $\omega^{\mathbf{si}}$ with each column of $R^{\mathbf{si}}$.

4. The location of a vehicle is described by its longitude λ from Greenwich, its latitude δ, and its altitude H above the surface of the earth as shown in Figure 1.18. Of course, the earth itself rotates at angular velocity $\omega_{\oplus}$ about its polar axis. A coordinate frame $\mathbf{s}$ has its origin at the center of the earth, with the $\mathbf{s}_1$ vector in the plane of the equator,

the $\mathbf{s}_2$ vector pointing at the vehicle, and $\mathbf{s}_3$ completing the right-handed set. Show that the angular velocity of the frame $\mathbf{s}$ is

$$\omega^{si} = \dot{\delta}\,\mathbf{s}_1 + (\omega_\oplus + \dot{\lambda})\sin\delta\,\mathbf{s}_2 + (\omega_\oplus + \dot{\lambda})\cos\delta\,\mathbf{s}_3 \tag{1.144}$$

and show that the inertial velocity of the vehicle is

$$\mathbf{v} = -(R+H)(\omega_\oplus + \dot{\lambda})\cos\delta\,\mathbf{s}_1 + \dot{H}\,\mathbf{s}_2 + (R+H)\dot{\delta}\,\mathbf{s}_3 \tag{1.145}$$

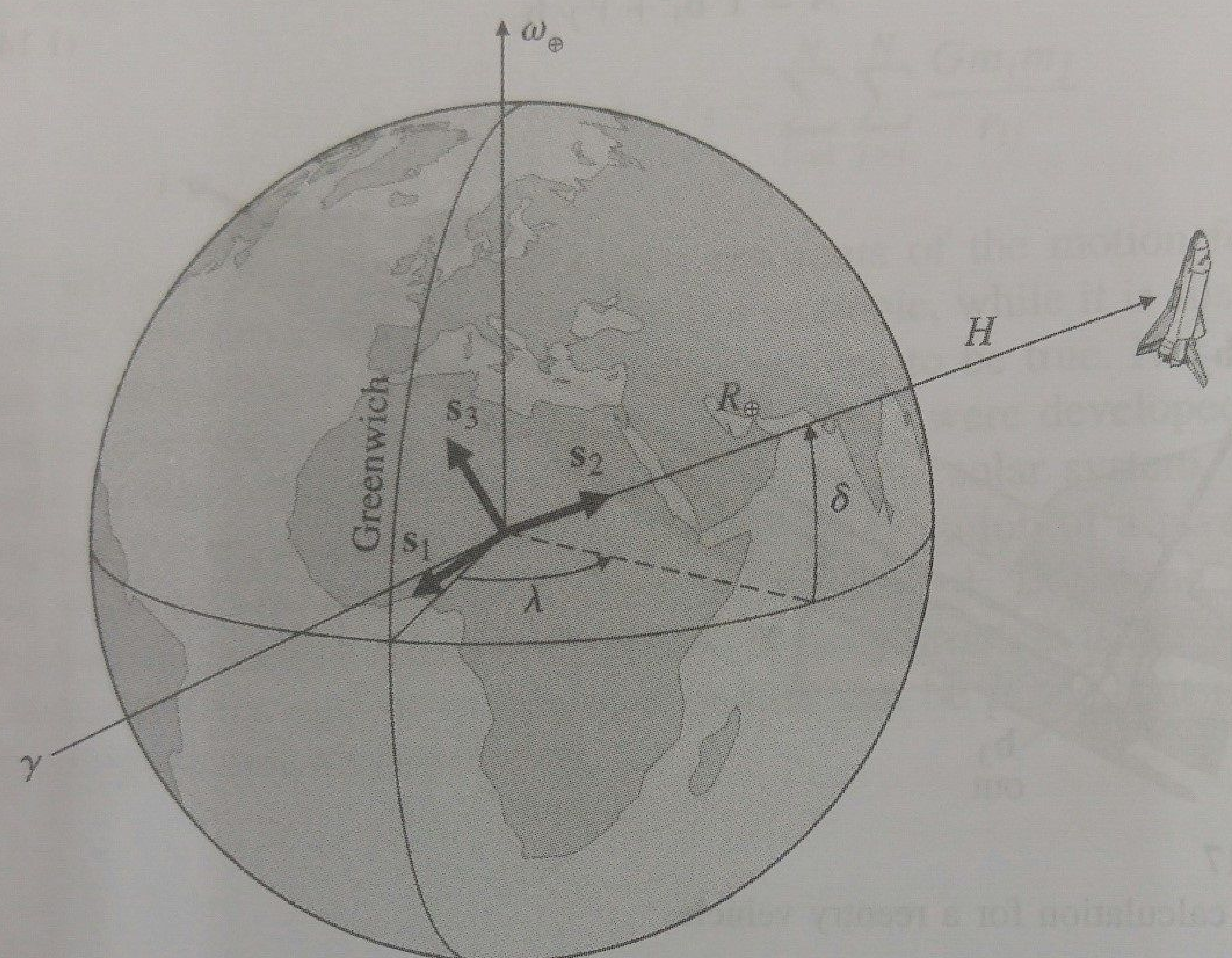

FIGURE 1.18
Flight vehicle located by altitude H, longitude λ, and latitude δ with respect to the rotating earth.

5. A future radio telescope might be constructed with a very large reflector and a separate spacecraft located at the focal point, a fixed distance R (perhaps several kilometers) from the main reflector. As the reflector changes its orientation, the two angles ϕ and θ shown in Figure 1.19 change with time. Write the angular velocity of the $\mathbf{s}$ frame shown in the figure and find an expression for the velocity the spacecraft needs to remain at the focal point.

6. An unfortunate astronaut, mass m, has lost his footing in the "low gravity" at the hub of the space station shown in Figure 1.20 and is about to "fall" down one of the spokes of the wheel. The station rotates at angular speed ω. Write equations of motion for the astronaut using coordinates x and y. On which side of the spoke should the ladder be located?

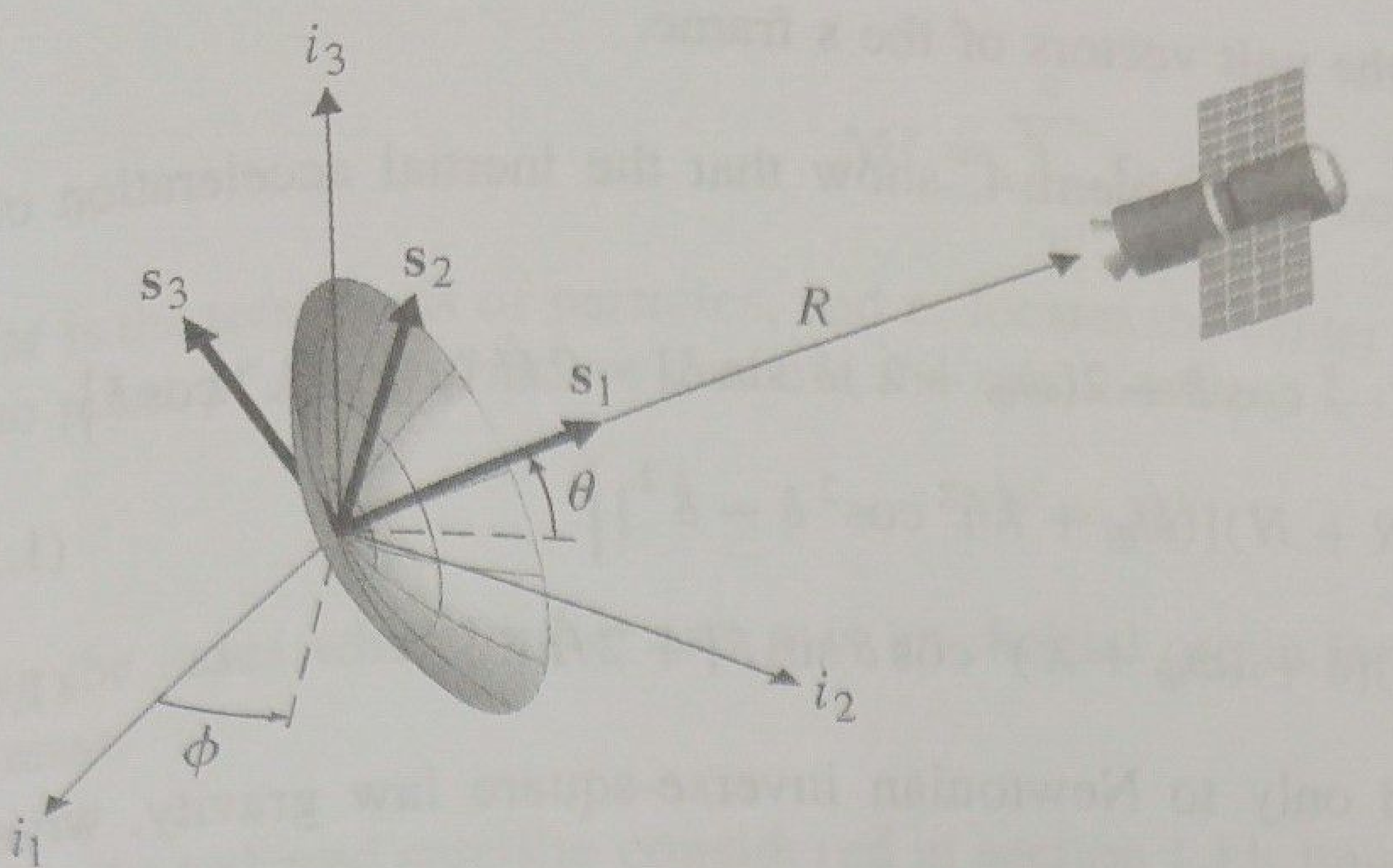

FIGURE 1.19
Receiver spacecraft stationkeeping at the focal point of a very large dish antenna.

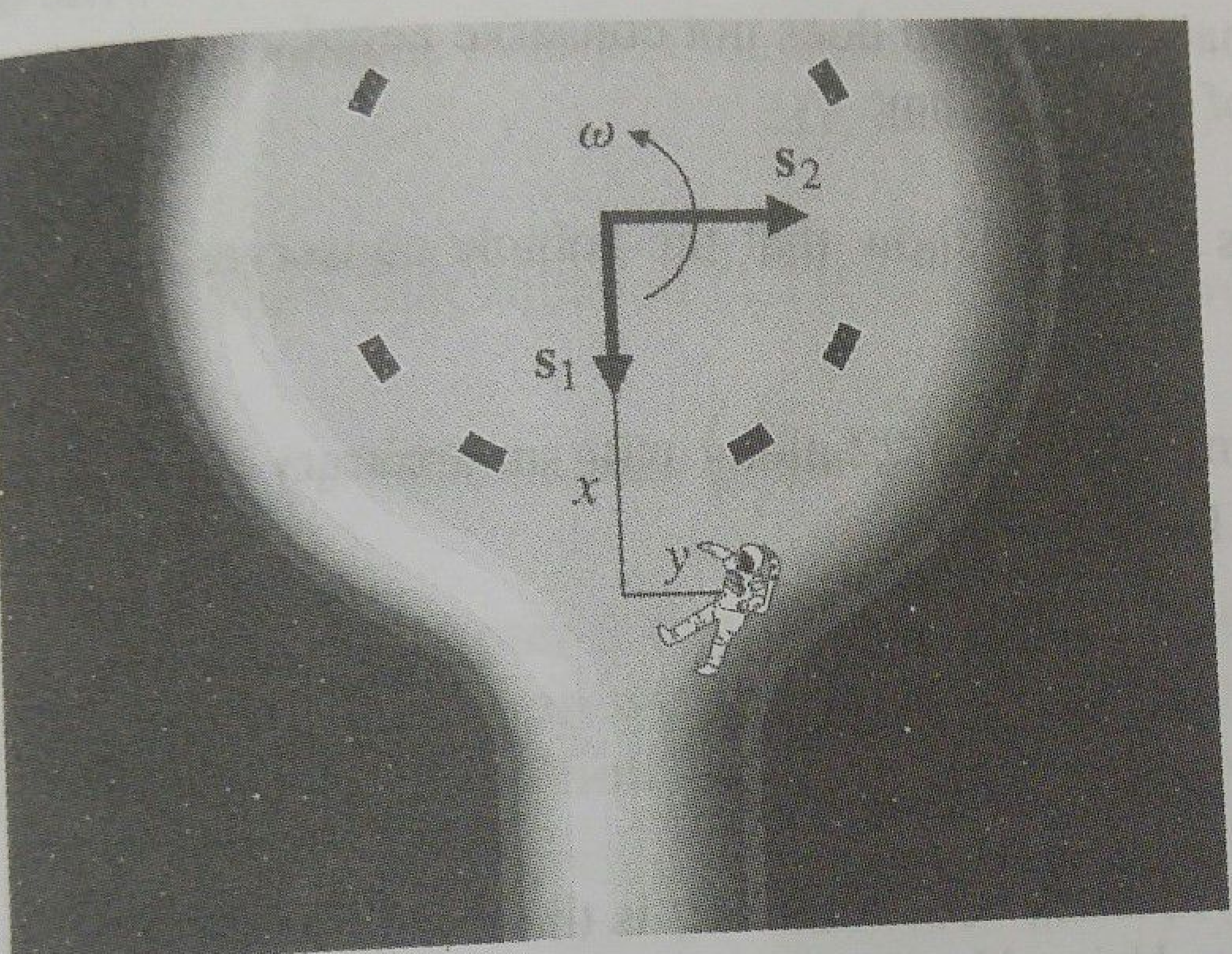

FIGURE 1.20
A "falling" astronaut in a rotating space station.

7. Rederive equations of motion for the baseball in the space colony considered in section 1.8. Use the cylindrical coordinates r, ϕ, z shown in Figure 1.21.

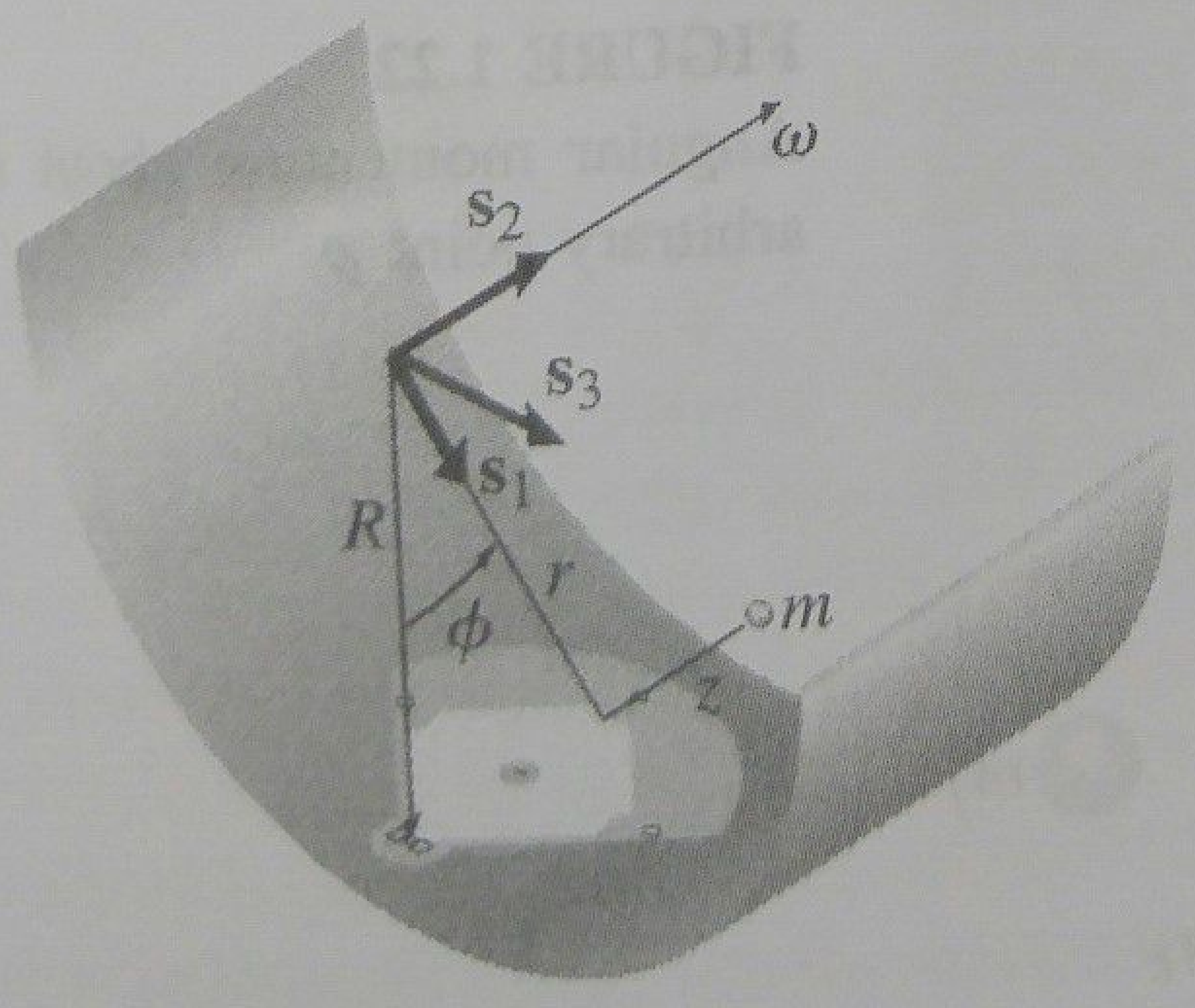

FIGURE 1.21
Polar coordinates for the baseball problem.

8. For the radio telescope considered in problem 5, find an expression for the acceleration of the receiver spacecraft in terms of the angles ϕ and θ and their first two derivatives

Express your answer in the unit vectors of the $\mathbf{s}$ frame.

9. For the vehicle considered in problem 4, show that the inertial acceleration of the spacecraft is

$$\mathbf{a} = \mathbf{s}_1 \left\{ (R+H)[-\ddot{\lambda}\cos\delta + 2(\omega_\oplus + \dot{\lambda})\dot{\delta}\sin\delta] - 2\dot{H}(\omega_\oplus + \dot{\lambda})\cos\delta \right\}$$

$$+ \mathbf{s}_2 \left\{ \ddot{H} - (R+H)[(\omega_\oplus + \dot{\lambda})^2\cos^2\delta - \dot{\delta}^2] \right\} \quad (1.146)$$

$$+ \mathbf{s}_3 \left\{ (R+H)[\ddot{\delta} + (\omega_\oplus + \dot{\lambda})^2\cos\delta\sin\delta] + 2\dot{H}\dot{\delta} \right\} \quad (1.147)$$

If the vehicle is subjected only to Newtonian inverse-square law gravity, write its equations of motion.

10. Show that the spherical pendulum of section 1.6 conserves energy, and write an expression for E. Also, show that this system does not conserve angular momentum, but that the vertical component of $\mathbf{H}$ *is* constant.

11. Using the results of problems 4 and 9, show that the vehicle conserves total energy. Write an expression for E.

12. A system of particles is shown in Figure 1.22 along with an arbitrary point p. Define the angular momentum about point p as

$$\mathbf{H}^p = \sum_{i=1}^{N} m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i \quad (1.148)$$

Calculate $\dot{\mathbf{H}}^p$, and substitute for $\mathbf{r}_i = \mathbf{R}_i - \mathbf{R}$, where $\mathbf{R}_i$ is the inertial position vector of particle i, and $\mathbf{R}$ locates point p. Using Newton's second law, show that this becomes

$$\dot{\mathbf{H}}^p = \mathbf{M}^p + \ddot{\mathbf{R}} \times M\mathbf{r}_c \quad (1.149)$$

where the torque about point p is

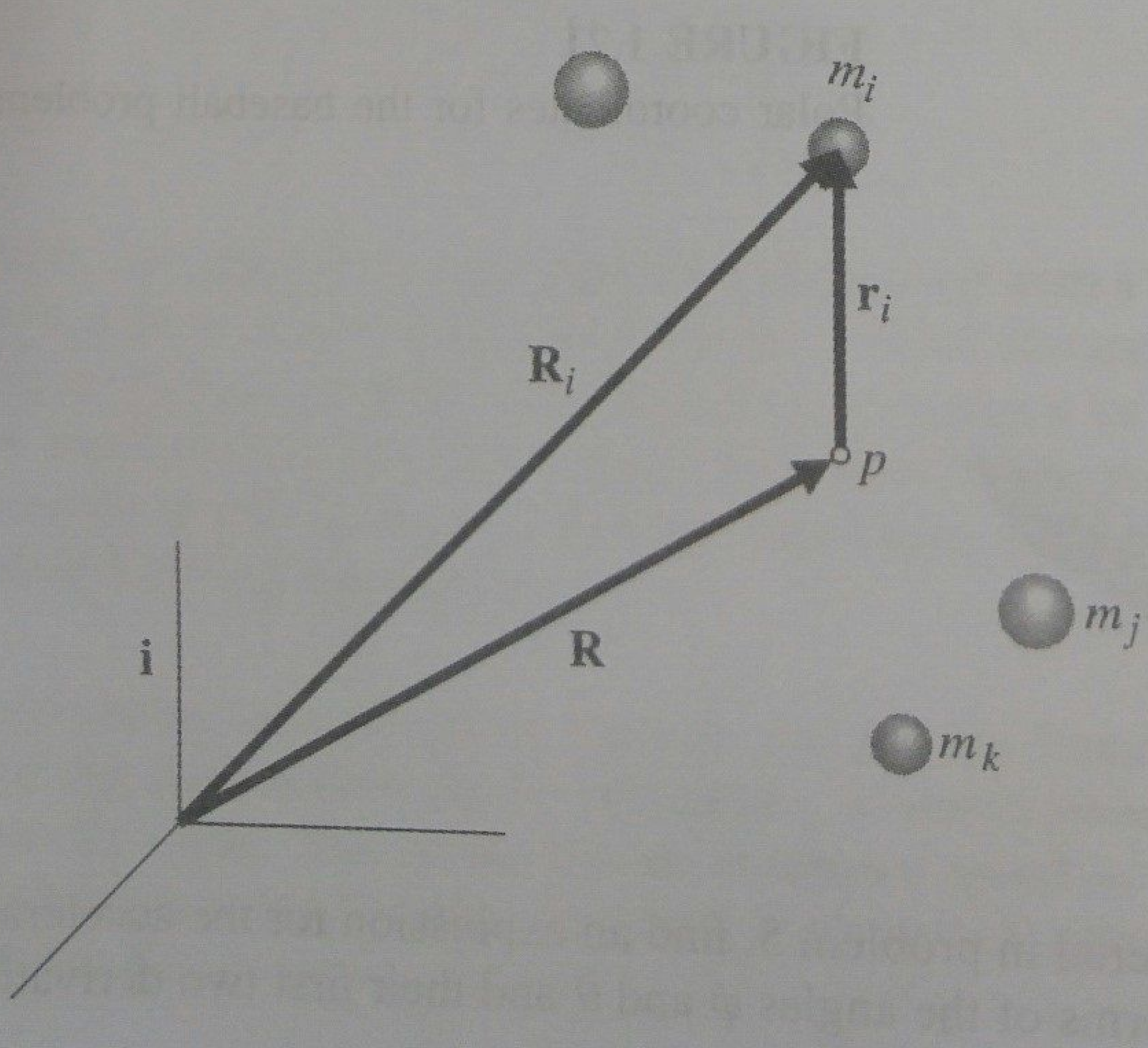

FIGURE 1.22
Angular momentum about an arbitrary point p.

$$\mathbf{M}^p = \sum_{i=1}^{N} \mathbf{r}_i \times \mathbf{F}_i \tag{1.150}$$

M is the total mass of particles, and $\mathbf{r}_c$ locates the center of mass of the system relative to point p. Argue that we can use the relationship

$$\mathbf{M}^p = \dot{\mathbf{H}}^p \tag{1.151}$$

in the cases where either (a) p is an inertial origin, or (b) p is the system center of mass.

13. For the tethered satellite considered in section 1.11, finish incorporating the assumption that the tether length δr is much less than the radius vectors $\mathbf{r}_1$ or $\mathbf{r}_2$. Use the binomial theorem to expand equation (1.114) to first order in $\delta\mathbf{r}$, and show it becomes

$$M\ddot{\mathbf{r}}_c = -\frac{GM_\oplus M}{r_c^3}\mathbf{r}_c \tag{1.152}$$

Similarly, show that equation (1.116) for the relative position becomes

$$\delta\ddot{\mathbf{r}} = GM_\oplus \left(\frac{\delta\mathbf{r}}{r_c^3} - \frac{3\mathbf{r}_c \cdot \delta\mathbf{r}}{r_c^5}\mathbf{r}_c \right) - T\,\frac{m_1 + m_2}{m_1 m_2}\frac{\delta\mathbf{r}}{|\delta\mathbf{r}|} \tag{1.153}$$

We shall see in Chapter 2 that the first equation states that the center of mass of the tether system follows an unperturbed two-body orbit. The first bracketed term in the relative equation of motion above is called the gravity-gradient force.